1 Solution — GIAM 6.1 Problem 3

1.1 Problem

The inverse of a relation R is denoted R^{-1}. It contains exactly the same ordered pairs as R but with the order switched:

$$R^{-1} = \{(b, a) : (a, b) \in R\}.$$

Define a relation S on $\mathbb{R} \times \mathbb{R}$ by

$$S = \{(x, y) : y = \sin x\}.$$

What is S^{-1}? Draw a single graph containing S and S^{-1}.

1.2 Computing S^{-1}

Apply the swap $(a, b) \mapsto (b, a)$ to every pair in S. Since

$$(a, b) \in S \iff b = \sin a,$$

the swapped pair (b, a) lies in S^{-1} exactly when this same equation holds. Renaming the coordinates of the swapped pair — let $x = b$ and $y = a$ — gives

$$S^{-1} \;=\; \{(x, y) \in \mathbb{R} \times \mathbb{R} \;:\; x = \sin y\}.$$

In words: S^{-1} is the set of all points whose first coordinate is the sine of the second.

A few matched pairs, $(a, b) \in S$ alongside the corresponding $(b, a) \in S^{-1}$:

- $(0, 0) \in S$ and $(0, 0) \in S^{-1}$.
- $(\pi/2,\, 1) \in S$ and $(1,\, \pi/2) \in S^{-1}$.
- $(\pi,\, 0) \in S$ and $(0,\, \pi) \in S^{-1}$.
- $(-\pi/2,\, -1) \in S$ and $(-1,\, -\pi/2) \in S^{-1}$.

1.3 A Single Graph

Swapping coordinates is exactly *reflection across the line* $y = x$: a point and its swap are mirror images through that diagonal. So S^{-1} is the reflection of S in $y = x$ — the horizontal sine wave $y = \sin x$ becomes a vertical sine wave $x = \sin y$.

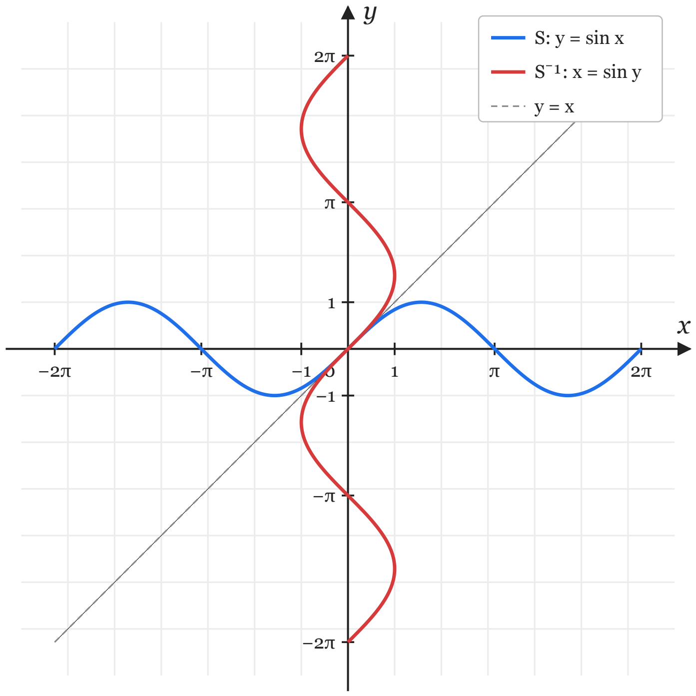

Figure 1.1: S (blue) and the inverse relation S⁻¹ (red) on the same axes; the dashed line y = x is the axis of symmetry.

The blue curve is S; the red curve is S^{-1}. The two are perfect mirror images across the dashed line $y = x$. Their only point of intersection is the origin, since $\sin x = x$ has no other real solutions.

1.4 Remarks

S^{-1} is a relation but not a function. The curve $x = \sin y$ fails the vertical line test: for any $c \in [-1, 1]$, the vertical line $x = c$ meets S^{-1} in *infinitely many* points, because

$$\{y \in \mathbb{R} : \sin y = c\} \;=\; \{\arcsin c + 2\pi k : k \in \mathbb{Z}\} \;\cup\; \{\pi - \arcsin c + 2\pi k : k \in \mathbb{Z}\}.$$

The familiar function $\arcsin : [-1, 1] \to [-\pi/2,\, \pi/2]$ is recovered by *restricting* S^{-1} to its **principal branch** — the portion of $x = \sin y$ lying in the horizontal strip $-\pi/2 \le y \le \pi/2$.

This illustrates a general pattern: inverting a non-injective function yields a multivalued relation, and a single-valued inverse function exists only after choosing a branch.